

Ansible introduction:-

Derops tasks before automation

1. have the server - onpremise, cloud, virtualization
2. Install the OS
3. Setup networking
4. Install web server (apache, nginx)
5. make changes into config file
6. copy some website data into the document root
7. service start

Ansible architecture:-

We do not need to install ansible on managed host because ansible is agent less.

2) Configure repo -

<https://repo.almaelinux.org/almaelinux/yum/AppStream/x26-64/os>

→ yum install -y ansible -*

→ dnf install <https://dl.fedoraproject.org/pub/epel/epel-release-latest-9.noarch.rpm> -y

1. pwd / ansible.cfg → preference highest
2. ~ / ansible.cfg
3. /etc / ansible

⇒ useradd admin

⇒ vim /etc / hosts

ip address	www.ndt.com	nd2
ip address	www.ndt.com	nd1

⇒ admin@ cd /ansible

vim ansible.cfg
[defaults]

inventory = /home/admin/ansible/hosts

→ vim hosts

nd1 or [dev] or [prod] or [web-serve: children]
nd2 or nd1 or mhl dev
nd2 or mhl prod

→ desired group name

2) ansible -m module -a attribute

→ two groups
already

2) ansible -m ping all -k

all
ungrouped

• ansible --version

• ansible -m copy -a "src = ./test.txt dest = /tmp" all -ls

• ansible -m command -a "ls -l /tmp/test.txt" -k all

↳ we can use raw, shell or command
↳ powerful

→ Exchange SSH keygen for avoid password.

ssh-copy-id nd1

ssh-copy-id nd2

⇒ vim ansible.cfg

[privilege-escalation]

become = true

become-method = sudo

become-user = root

become-ex-pass = false

2) " path = /etc state = directory "

Playbook :-

vim example.yaml

- name: create the group as devops

hosts: all

tasks:

- name: group creation

group:

name: devops

state: present

- name: create user
user:
name: dev
group: devops

- name: copy data.txt
copy:
src: ./data.txt
dest: /home/dev/

- name: this is the updated playbook

hosts: ndt

tasks:

- name: edit data.txt

lineinfile:

path: /home/dev/data.txt

line: Hello welcome to bigdata

- ansible-playbook example.yaml --syntax-check
ansible-playbook example.yaml

- vim fstab.yaml

- name: modify fstab file

hosts: all

tasks:

- name: fstab

lineinfile:

path: /etc/fstab

line: /dev/sr0 /mnt iso9660 defaults 0 0

- name: mount

shell: mount -a

- ansible-playbook fstab.yaml

stepo configuration :-

```
---
- name: confg stepo
  hosts: all
  tasks:
    - name: confg appstream repo
      yum_repository:
        name: Appstream
        description: This is App Repo
        baseurl: file:///mnt/Appstream
        file: Appstream
        gpgcheck: no
        enabled: yes
    - name: confg baseos repo
      yum_repository:
        name: BaseOS
        description: This is App Repo
        baseurl: file:///mnt/BaseOS
        file: BaseOS
        gpgcheck: no
        enabled: yes
- ansible-playbook yum_stepo.yaml
```

web-server.yaml :-

```
---
- name: configure web-server
  hosts: all
  tasks:
    - name: install apache
      yum:
        name: httpd
        state: present
    - name: manage apache service
      service:
        name: httpd
        state: started
        enabled: yes
```

- name: manage firewall
firewalld:
port: 80/tcp
permanent: true
state: enabled
immediate: true

- name: copy web-content
copy:
src: ./index.html
dest: /var/www/html

- name: restart httpd service
service:
name: httpd
state: restarted

Task 7. yaml :-

```
---
- name: Task 1
  hosts: all
  vars_files:
    - ./users_vars.yaml
```

tasks:

- name: creating group named {{ grp-name }}

```
group:
  name: "{{ grp-name }}"
  state: present
```

- name: Creating user named {{ user-name }} with primary group {{ group-name }}

```
user:
  name: "{{ user-name }}"
  group: "{{ group-name }}"
  password: "{{ user-name-password | password_hash('sha512') }}"
```

- name: Creating users named {{ sec-user }} with primary group {{ group-name }}

```
user:
  name: "{{ sec-user }}"
  groups: "{{ group-name }}"
```

```
password: "{{sec-user-password | password_hash('sha512')}}"
append: yes
```

multi-user.yml :-

```
---
- name: creating multi
  hosts: all
  tasks:
    - name: creating multi user
      group:
        name: "{{ item }}"
      loop:
        - group1
        - group2
    - name: users
      user:
        name: "{{ item.user }}"
        group: "{{ item.group }}"
      loop:
        - user: user1
          group: group1
        - user: user2
          group: group2
```

ansible.cfg :-

[defaults]

inventory = /home/admin/ansible/inventory

remote_user = admin

ask_vault_pass = true

ask_pass = true

[privilege_escalation]

become = true

become_method = sudo

become_user = root

become_ask_pass = false

vars.yml :-

App: apache
package: httpd
service: httpd.service
port: 80/tcp
DC: /var/www/html

users-var.yml

group-name: Asig
user-name: Intra
user-name-password: gredhat
sec-user: UK
sec-user-password: gcut

print.yml

- name: print

hosts: all

vars:

name: pankaj

job: student

tasks:

- name: print

debug:

msg: Hello my name is {{name}} and I am a {{job}}

- name: printing fqdn

debug:

msg: the fqdn is the machine is {{ansible_processor[1]}}
and the dns servers for it is {{ansible_dns.
nameservers[0]}}

→ ansible -m command -u "mkdir -p /etc/ansible/facts.d" all

→ vim custom-facts-fact

[fun]

kiwi = fruit

matrix = movie

octopus = '8 legs'

→ ansible -m copy -a "src = ./custom-facts-fact dest = /etc/ansible/
facts.d/ owner = admin" all

lvm.yum:-

- name: lvm creation

hosts: all

tasks:

- name: vg not available

~~when:~~

debug:

msg: vg is not available

~~when:~~

when: ansible_lvm.vgs.myvg is not defined

- name: lvm create of 1500m

lvol:

vg: myvg

lv: mylv

size: 1500m

when: ansible_lvm.vgs.myvg.free_g >= "1.6"

- name: print msg size not available

debug:

msg: size is not available

when: ansible_lvm.vgs.myvg.free_g >= "1.6"

- name: lvm create of 800m

lvol:

vg: myvg

lv: mylv

size: 800m

when: ansible_lvm.vgs.myvg.free_g <= "1.6"

- name: format lv

filesystem:

fstype: ext4

dev: /dev/myvg/mylv

partition.yml

- name: create partition

hosts: all

tasks:

- name: print a msg device not available

debug:

msg: device is not available

when: ansible_devices.sdb is not defined

- name: create 3G partition

parted:

device: /dev/sdb

number: 1

state: present

part-end: 3GiB

when: ansible_devices.sdb.size > "3g"

- name: print a message size not available

debug:

msg: 3g size is not available

when: ansible_devices.sdb.size < "3g"

- name: Create 1.8G partition

parted:

device: /dev/sdb

number: 1

state: present

part-end: 1.8 GiB

when: ansible_devices.sdb.size < "3g"

- name: format partition

filesystem:

fstype: ext4

dev: /dev/sdb1

- name: mount

mount:

path: /data

src: /dev/sdb1

fstype: ext4

state: present

fetch.yaml

```
---
- name: fetch module use
  hosts: all
  vars:
    - name: fetch
      fetch:
        src: /etc/fstab
```

values.j2

```
{% for host in ansible_processor %}
{% host %}
{% endfor %}
```

hw-loop.yaml

```
---
- name: hello
  hosts: all
  vars:
    values:
      - abc
      - xyz
      - pqr

  tasks:
    - name: copy a file
      template:
        src: ./values.j2
        dest: /opt/values.j2
```

```
- oldvalue: biosversion
  newvalue: "{{ ansible_bios_version }}"

- oldvalue: memory
  newvalue: "{{ ansible_memtotal_mb }}"

- oldvalue: sda_size
  newvalue: "{{ ansible_devices.sda.size }}"

- oldvalue: sdb_size
  newvalue: "{{ ansible_devices.sdb.size }}"
```

hardware.txt:-

```
INVENTORY-HOSTNAME = hostname
BIOS-VERSION = biosversion
TOTAL-MEMORY = memory
SDA-SIZE = sda_size
SDB-SIZE = sdb_size
```

hard.yaml

```
---
- name: hardware script
  hosts: all
  tasks:
    - name: copy a file
      copy:
        src: ./hardware.txt
        dest: /root/hardware.txt

    - name: use replace module
      replace:
        path: /root/hardware.txt
        regexp: "{{ item.oldvalue }}"
        replace: "{{ item.newvalue }}"

    loop:
      - oldvalue: hostname
        newvalue: "{{ ansible_hostname | default('NONE') }}"
```

mkdir roles

ansible-galaxy init apache

cd tasks - webserver.yml

cd handlers - handlers

cp index.html /home/admin/ansible/roles/~~ht~~apache/files/

cp picat.jpg /home/admin/ansible/roles/apache/templates/

cd roles - vim main.yml

vim ansible.cfg

roles-path = /home/admin/ansible/roles